Triangle singularity

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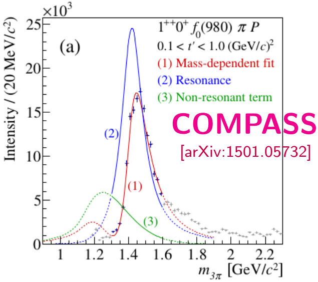
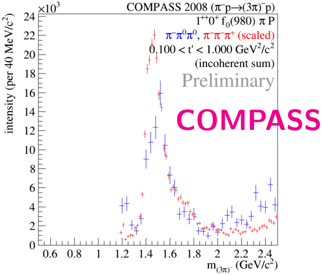
The phenomenon has been studied since 1960-th and known with different names:

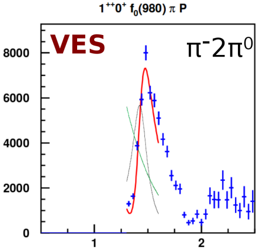
- modified inelastic Peierls machanism,
- triangle machanism,
- triangle singularity
- first order correction to isobar model (Aitchison-Brehm),
- first iteration on Khuri and Treiman equation .

New a_1 - $1^{++}0^+ f_0(980)\pi\ P-$wave

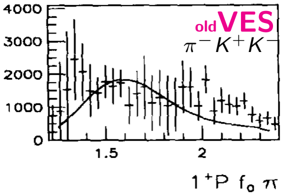




[PoS **Hadron2013**, 087 (2013)]



[PoS **Hadron2013**, 088 (2014)]

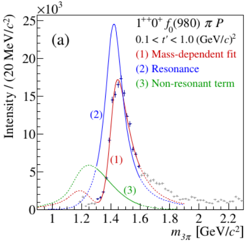


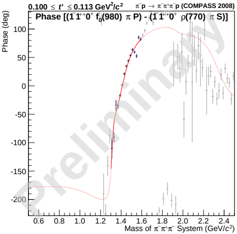
[Berdnikov *et al.*, Nuovo Cim. 107, 1941 (1994)]

Is $a_1(1420)$ a resonance?

Might be, because

- narrow bump,
- sharp phase motion

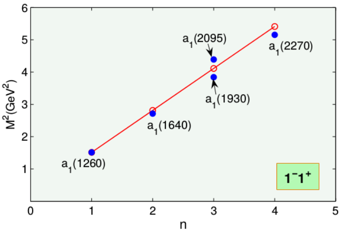




Issues to be clarified:

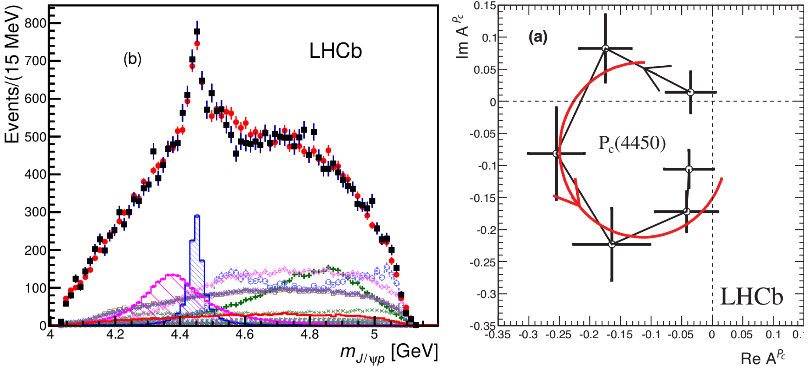
- $a_1(1420)$ doesn't fit to radial excitation trajectory as well as Regge-trajectory.
- Closeness of $a_1(1260)$.
- Width is too small compared to $a_1(1260)$.
- Strange coincidence of the mass to $K^\star(892)\bar{K}$ threshold, $\approx 1.380\,\mathrm{GeV}/c^2$.

[arXiv:1401.1134]



Dalitz plot singularity: the pentaquark $P_c(4450)$

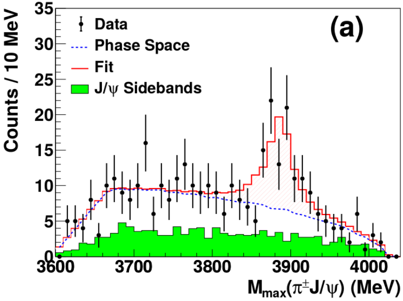
$$\Lambda_b \to K\,(J/\Psi\, p)$$

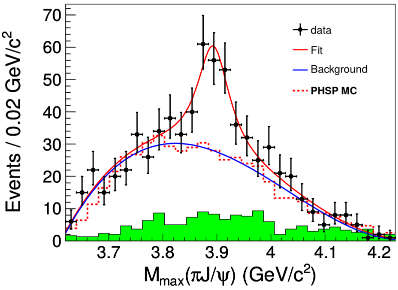


- The minimal quark content is $uudc\bar{c}$,
- The phase

Dalitz plot singularity: $Z^{\pm}(3900)$-state

$$Y(4260) \to (J/\Psi\pi)\pi, \qquad \Psi'(4160) \to (J/\Psi\pi)\pi$$





- The state lies at charmonium sector, has $c\bar{c}$ quarks.
- It is charged, then the minimal quark content is $udc\bar{c}$.

Contents

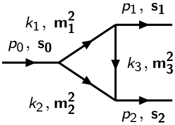
What is the triangle singularity?

Feynman approach

$$A(s_0, s_1, s_2) = g^3 \int \frac{d^4 k_1}{(2\pi)^4 i} \frac{1}{\Delta_1 \Delta_2 \Delta_3} = \frac{g^3}{16\pi^2} \int_0^1 \frac{\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z}{D} \delta(1 - x - y - z),$$

$$\Delta_i = m_i^2 - k_i^2, \quad D = x\,m_1^2 + y\,m_2^2 + z\,m_3^2 - x\,y\,s_0 - z\,x\,s_1 - y\,z\,s_2.$$

Positions of singularities are given by equations: [Landau, Nucl. Phys. **13**, 181 (1959)]



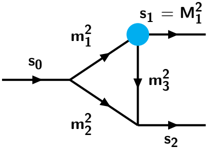
$$\begin{cases} k_i^2 = m_i^2, & i = 1 \ldots 3, \\ x\,k_{1\mu} - y\,k_{2\mu} + z\,k_{3\mu} = 0, & x, y, z \in [0, 1], \\ x + y + z = 1, & \end{cases}$$

- leading singularity: log. branching point, $A \sim \log(s - s_{\text{sing}})$,
- leading singularity: log. branching point, $A \sim \log(s - s_{\text{sing}})$,
- square root branching points, $A \sim \sqrt{s - s_{\text{sing}}}$ (normal thresholds).

The different kinematical regions, [Eden *et al.*, Cambridge Univ.Pr.(2002)]

Introduce normalized invariants

$$\begin{aligned} y_0 &= (s_0 - m_1^2 - m_2^2)/(2m_1 m_2), \\ y_1 &= (s_1 - m_1^2 - m_3^2)/(2m_1 m_3), \\ y_2 &= (s_2 - m_2^2 - m_3^2)/(2m_2 m_3) \end{aligned}$$



To simplify representation, we fix internal masses and $\mathbf{s_1}$.
Depending on value of $s_1(y_1)$, we have:

$|y_1| < 1$ "No decays": $M_1 < m_1 + m_3$, $m_1 < M_1 + m_3$, $m_3 < M_1 + m_1$.

- The triangle singularity is known as **anomalous threshold**. It sits below the normal threshold for s_0, s_2.
- Appears in form-factors.

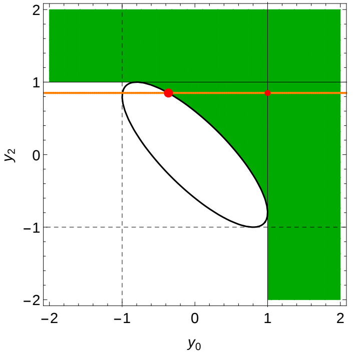
$y_1 < -1$ "Internal decay": $m_1 > M_1 + m_3$ or $m_3 > M_1 + m_1$.

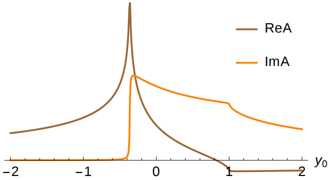
- Position is given by decay kinematics.
- Appears in decays.

$y_1 > 1$ "External decay": $M_1 > m_1 + m_3$.

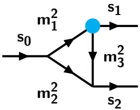
Is covered above, just if we choose another variable to fix.

"No decays" region: position of singularities





Invariant s_1 is fixed,
$|y_1| = |(s_1 - m_1^2 - m_3^2)/(2m_1 m_3)| < 1$.



- the position of leading log singularity is arc of ellipse,
- solid $y_0 = +1$, $y_2 = +1$ lines are normal thresholds: $(m_1 + m_2)^2$ for s_0 and $(m_2 + m_3)^2$ for s_2 (lower order singularities).
- Green area: $\mathrm{Im}A \neq 0$ (above threshold).

An example: deuteron size [Gribov lecture, Cambridge Univ.Pr.(2009)]

Non-relativistic quantum mechanics

Proton and neutron bound state,

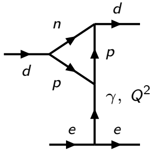
- binding energy: $\varepsilon = M_D - m_p - m_n$,
- ψ is the wave function, $\psi \sim e^{-r\sqrt{\varepsilon m}}$.
- Electron scattering amplitude: $f(q) = \frac{e^2}{q^2}F(q)$, $F(q)$ is a FF.
- $F(q)$ is given by charge density inside of the deuteron

$F(q) = \int d^3\mathbf{r}\, \psi^2(r)\, e^{i\mathbf{qr}} \approx 1 - \frac{q^2R^2}{2} + O(q^4)$
R is electromagnetic radius,

$$R^2 \sim (\varepsilon m)^{-1}$$

Relativistic-theory framework

Radius is determined by t-channel singularities.



The singularities in Q^2:

- Normal threshold $Q_0^2 = (2m_p)^2$
- Anomalous threshold $Q_0^2 \approx 16\varepsilon m$

Further examples

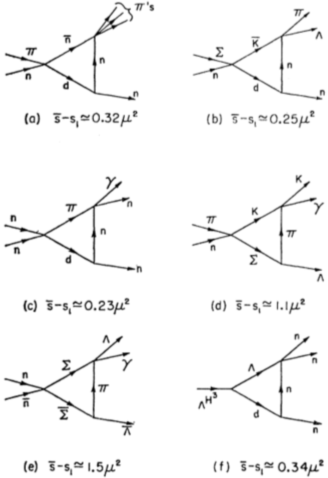
Selected examples:

- Mandelstam example $pp \to \pi\pi$.
- Some Experimental Consequences of Triangle Singularities in Production Processes

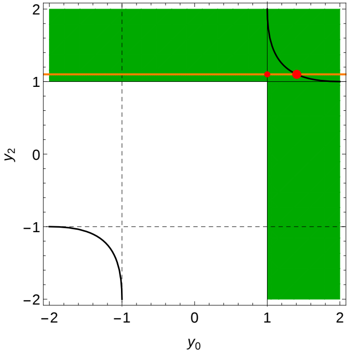
  [Landshoff *et al*, Phys. Rev. 127, 649 (1962)]
- The Deuteron as a composite two nucleon system

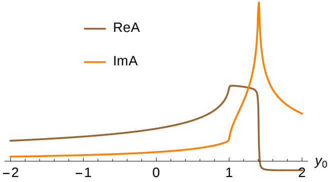
  [Anisovich *et al*, Nucl.Phys. A544 (1992) 747-792]
- Form-factors of meson decays

  [Melikhov, Phys.Rev. D53 (1996) 2460-2479]

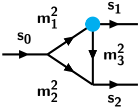


"Internal decay" region: position of singularities





Invariant s_1 is fixed,
$y_1 = (s_1 - m_1^2 - m_3^2)/(2m_1 m_3) < -1.$



- position of leading log singularity is solid arc of hyperbola,
- solid $y_0 = +1$, $y_2 = +1$ lines are normal thresholds: $(m_1 + m_2)^2$ for s_0 and $(m_2 + m_3)^2$ for s_2.
- Green area: $\mathrm{Im}A \neq 0$ (above threshold).

Examples

Selected examples:

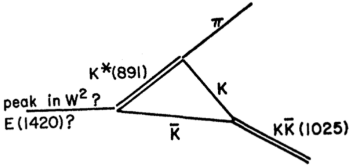
- The nature of C-meson. [Achasov *et al.*, Phys.Lett. B207 (1988) 199]
  $\rho(1450) \to (K^\star \bar{K} \to \pi K \bar{K}) \to \pi\phi$
- $\eta(1405/1475)$ puzzle.
  $\eta \to (K^\star \bar{K} \to \pi K \bar{K}) \to \pi a_0(980), \pi f_0(980)$,
  $f_1(1420) \to (K^\star \bar{K} \to \pi K \bar{K}) \to \pi a_0(980), \pi f_0(980)$.
  [Jia-Jun Wu *et al.*, Phys.Rev.Lett. 108 (2012) 081803],
  [Xiao-Gang Wu *et al.*, Phys.Rev. D87 (2013) 1, 014023]
- Some of XYZ states. [Szczepaniak, arXiv:1501.01691]
  $Z_c(3900) - \pi \to (D^\star \bar{D} \to \pi D \bar{D}) \to J/\Psi\pi$
  $\Upsilon(5s) - \pi \to (B^\star \bar{B} \to \pi B \bar{B}) \to \Upsilon(1s)\pi$
- The nature of $a_1(1420)$. [Mikhasenko *et al.*, arXiv:1501.07023]
  $a_1(1260) \to (K^\star \bar{K} \to \pi K \bar{K}) \to \pi f_0(980)$

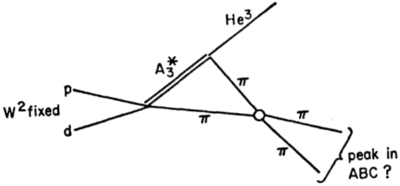
Schmid theorem [Schmid, Phys.Lett. (1966)]

Final-State Interactions and the Simulation of Resonances*†

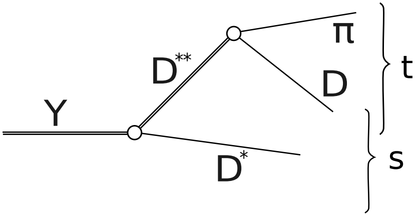
CHRISTOPH SCHMID
California Institute of Technology, Pasadena, California
(Received 25 July 1966)

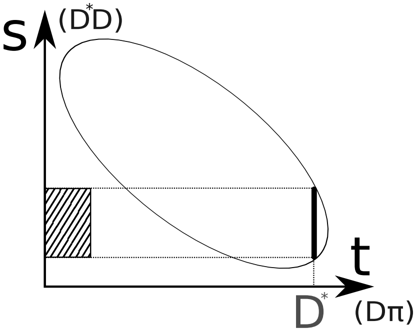
It is shown that the singularities of the triangle or rescattering diagram cannot produce peaks in the total transition rate. Rescattering bands can be seen on the Dalitz plot, but their projections into invariant-mass plots do not produce peaks, if all events on the Dalitz plot are included. In other words, the Peierls mechanism and its recent variants do not work, even if the singularity is on the physical boundary. The reason is a cancellation of the peak, which occurs when the diagram without rescattering is added coherently to the triangle diagram. It is also shown that interference of overlapping resonance bands does not produce the previously expected peaks in the total transition rate or in the two-body mass plots in three different cases. This result holds to first order in the ratio of the resonance width to the diameter of the Dalitz plot.



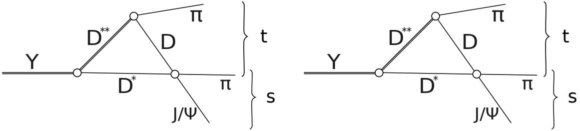


The example of $D^{\star}D$



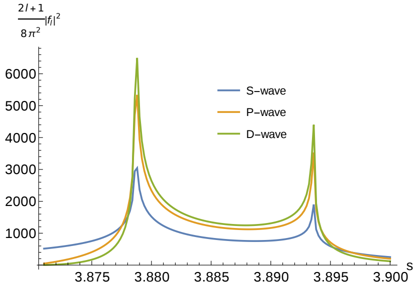


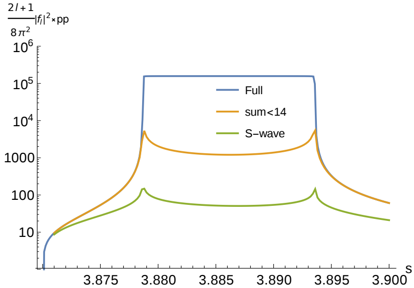
One has to add the triangle diagram and the projection from s-channel.



Projection of t-channel isopbar to s-channel

$$\frac{\mathrm{d}N}{\mathrm{d}s} \sim p_1 p_3 \int \frac{\mathrm{d}z_t}{\left|m_1^2 - t(z_t) - i\epsilon\right|^2},$$



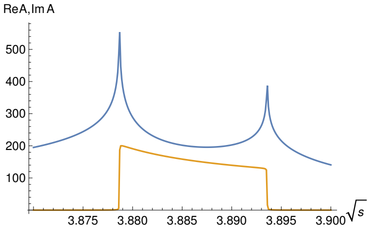


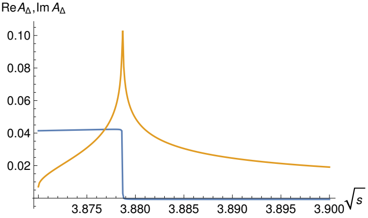
We expand $A(t(z_t), s)$ in the Legendre polynomials:

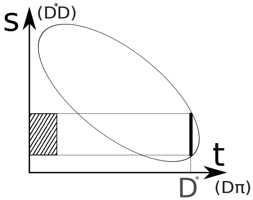
$$A(t(z_t), s) = \sum \frac{2l+1}{4\pi} f_l(s) P_l(z_t), \quad \int |A(s,t)|^2 \mathrm{d}z_t = \sum \frac{2l+1}{8\pi^2} |f_l(s)|^2 .$$

Sum

$$A = A_{\text{proj}} + A_\Delta T_{23} \to A_{\text{proj}}(1 + 2i\rho_2 T_{23}) = A_{\text{proj}} e^{2i\delta_{23}}.$$







The events on dalitz plot migrate along the X. In case of elastic $2 \to 2$ rescattering the triangle causes redistribution of events on dalitz plot.

In presence of inelastisity [A. Szczepaniak, ArXiv:1510.00695]

In presence of several channels redisctribution of events happens between different channels.

The explicit relalation was written for the case of two channels.

$$A_1(s,t,u) = \frac{1}{4\pi}[S_{0,11}(s) - 1]b_{0,1}(s) + A_1^t(t)$$

$$A_2(s,t,u) = \frac{1}{4\pi}\sqrt{\frac{\rho_1(s)}{\rho_2(s)}}S_{0,21}(s)b_{0,1}(s) + A_2^t(t)$$

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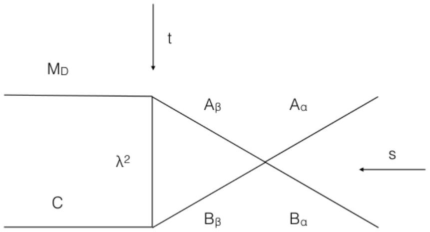


If we fix final state and integrate dalitz plot the leak/benefit of events shows up as an enhancement on 3-particle mass distribution

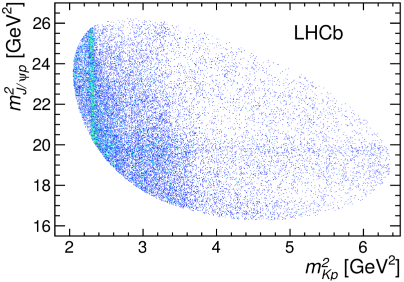
Local conclusion-I

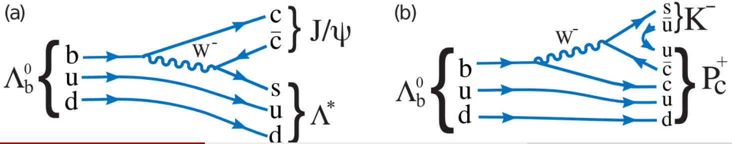
- Schmid cancelation is valid only for an elastic case.
- In presence of inelastisity the effect of a triangle diagram appears **in projections** for fixed total inveriant mass and **in 3-particle mass distribution** for fixed final state.
- The analytical relations for amplitude in the limit of triangle singularity can be written.
- $Z(3900)$ at $J/\Psi\,\pi$ is very good candidate to test the approach:
  - The peak is observed in several reactions ($\Psi'(4160) \to J/\Psi\pi\pi$ and $Y(4260) \to J/\Psi\pi\pi$),
  - $Y(4260) \to D^\star D\pi$ data are available.

LHCb analysis and observation of $J/\Psi\, p$ peak

The mass-dependent analysis has been done. Several isobar $\Lambda(\Sigma)$ with different LS couplings (up to 6) was used in $K\,p$ channel and the model are not able to reproduce the peak in $J/\Psi\, p$ projection.

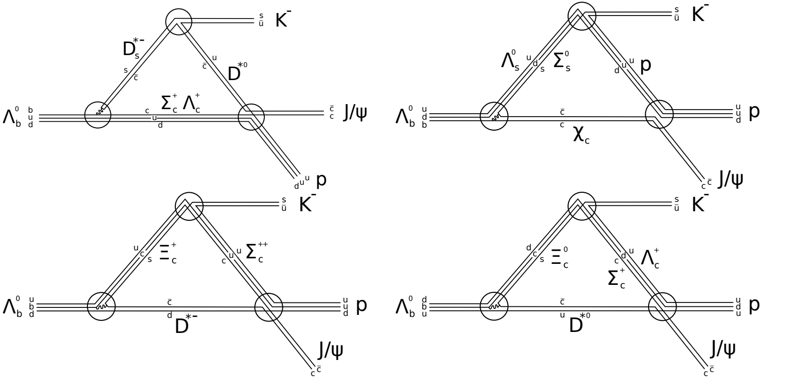
[LHCb collaboration, Phys. Lett. (2015)]



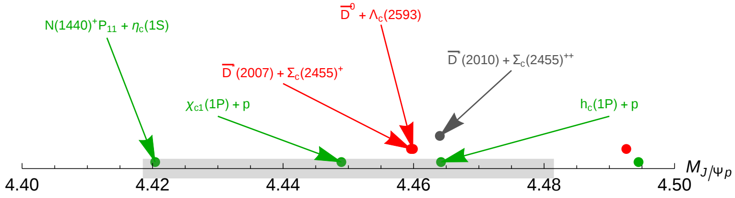


Possible triangle processes [M. Mikhasenko, arXiv: (2015)]

$$\Lambda_b^0 \to \begin{array}{rcl} D^0(u\bar{c}) & + & (dsc)\,\Xi^0 \\ D^-(d\bar{c}) & + & (usc)\,\Xi^+ \\ D_s^-(s\bar{c}) & + & (udc)\Sigma_c^+,\Lambda_c^+ \\ \Psi(c\bar{c}) & + & (uds)\Sigma^0,\Lambda^0 \end{array} \to K^-(s\bar{u}) + \begin{array}{rcl} D^0(u\bar{c}) & + & (udc)\Sigma_c^+,\Lambda_c^+ \\ D^-(d\bar{c}) & + & (uuc)\,\Sigma_c^{++} \\ D^0(u\bar{c}) & + & (udc)\Sigma_c^+,\Lambda_c^+ \\ \Psi(c\bar{c}) & + & (uud)\,N^+ \end{array} \to K^- + J/\Psi + p$$

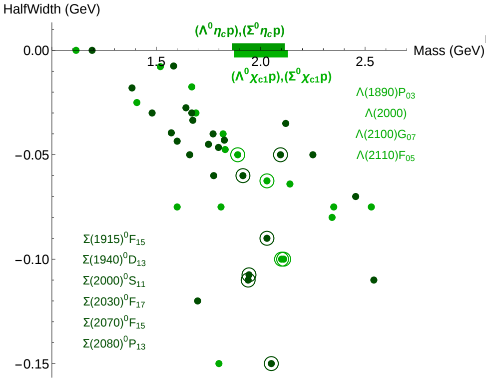


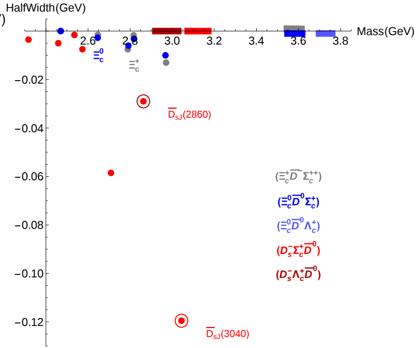
Relevant thresholds



- $\Lambda_c(2593) + \bar{D}^0$.
- $\Sigma_c^+(2455) + \bar{D}^{\star 0}(2007)$, suppressed at Λ_b vertex by isospin.
- $\Sigma_c^{++}(2455) + \bar{D}^{\star -}(2010)$.
- $p + \chi_{c1}$, suppressed because of the parity of $c\bar{c}$ state is changed in gluon exchange.
- $p + h_c$, suppressed by C-parity.

Third particle



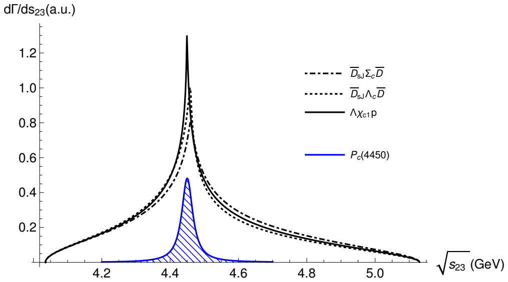


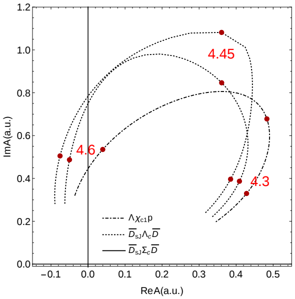
Interesting to look at the amplitude for

- $\Lambda_b \to D_{sJ}(2860)\,\Lambda_c(2593) \to K^-\,\bar{D}^0\,\Lambda_c \to K^-\,J/\Psi\,p,$
- $\Lambda_b \to D_{sJ}(3040)\,\Sigma_c^+(2455) \to K^-\,\bar{D}^{\star 0}(2007)\,\Sigma_c^+ \to K^-\,J/\Psi\,p,$
- $\Lambda_b \to \Lambda(1890)\,\chi_{c1} \to K^-\,p\,\chi_{c1} \to K^-\,J/\Psi\,p.$

Mass distribution

$$\frac{\mathrm{d}\Gamma_{\Lambda_b \to K^- J/\Psi p}}{\mathrm{d}s_{23}} = \frac{1}{2\sqrt{s_0}} \left| A_{\Lambda_b \to K^- J/\Psi p} \right|^2 \frac{\mathrm{d}\Phi_3}{\mathrm{d}s_{23}}$$



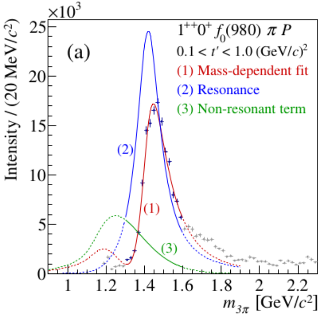


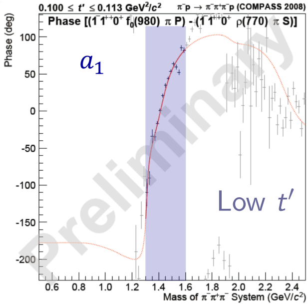
- $\Lambda_b \to D_{sJ}(2860)\,\Lambda_c(2593) \to K^-\,\bar{D}^0\,\Lambda_c \to K^-\,J/\Psi\,p$,
- $\Lambda_b \to D_{sJ}(3040)\,\Sigma_c^+(2455) \to K^-\,\bar{D}^{\star 0}(2007)\,\Sigma_c^+ \to K^-\,J/\Psi\,p$,
- $\Lambda_b \to \Lambda(1890)\,\chi_{c1} \to K^-\,p\,\chi_{c1} \to K^-\,J/\Psi\,p$.

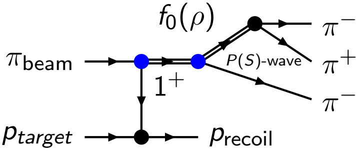
Local conclusion-III

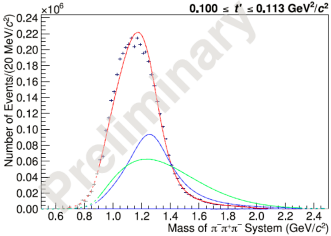
- The idea of the pentaquark existance is very attractive, the peak is seen in the row spectrum.
- There are many threshold in $J/\Psi p$ channels. They are not taken into account.
- The simple isobar model doesn't work there.
- Based on the knowlegde how the triangle correction appear we can select relevant processes. Several of them indeed produce the peaks with phase motion.
-

1^{++}-waves in COMPASS: $f_0\pi$ P-wave and $\rho\pi$ S-wave



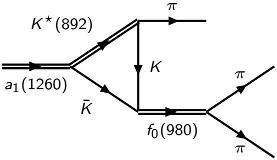






The triangle diagram

- condition of "internal decay" is satisfied: $m_{K^\star} > m_K + m_\pi$
- $a_1(1260)$ is rather wide, so s_0 can go above $K^\star \bar{K}$ threshold.
- f_0 sits at the $K\bar{K}$ threshold.

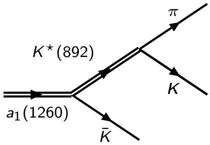


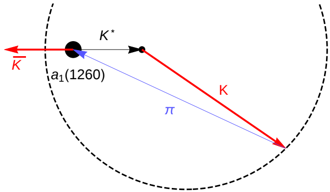
The triangle singularity can appear in the physical region. Whether it does is given by Landau equations.

Two isospin combinations contribute:

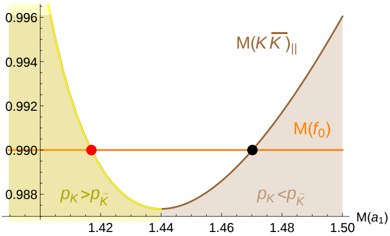
- $a_1^-(1260) \to K^{\star 0} K^- \to \pi^- K^+ K^- \to \pi f_0(980)$
- $a_1^-(1260) \to K^{\star -} K^0 \to \pi^- K^0 \bar{K}^0 \to \pi f_0(980)$

Kinematical formulation of Landau equations





Imagine cascade reaction $a_1(1260) \to K^\star(892)\bar{K}$, then $K^\star \to K\pi$, and calculate invariant mass of K and $\bar{K}$ for the case when K is parallel to $\bar{K}$.



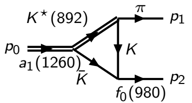
Particular form of Landau conditions:

- All particles in loop are on mass shell.
- The alignment of moments $\vec{p}_K \uparrow\uparrow \vec{p}_{\bar{K}}$.
- K is faster then $\bar{K}$.

Calculation of the rescattering: $a_1(1260) \to K^\star \bar{K} \to f_0 \pi$

For the realistic decay, we have expression in the numerator

$$\mathbb{M}^{(\mathrm{vpp})}_{a_1 \to \pi f_0} = \left[\quad \right]$$

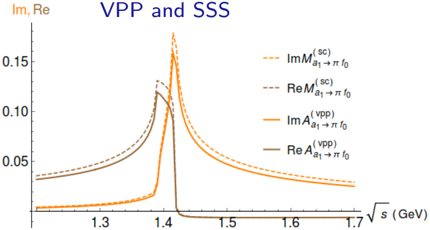


- Spin-Parity of particles.
- Width of $K^\star$

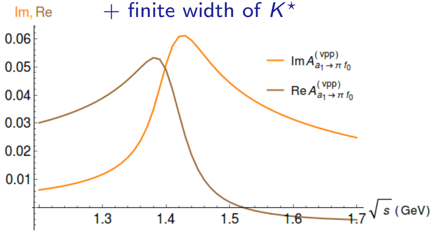
$$\mathbb{M}^{(\mathrm{vpp})}_{a_1 \to \pi f_0} = g^3 \int \frac{\mathrm{d}k_1^4}{(2\pi)^4\, i} \frac{\varepsilon_{0\mu} \left(g^{\mu\nu} - \frac{k_1^\mu k_1^\nu}{k_1^2} \right) (p_1 - k_3)_\nu}{(m_1^2 - k_1^2 - i\epsilon)(m_2^2 - (p_0 - k_1)^2 - i\epsilon)(m_3^2 - (k_1 - p_1)^2 - i\epsilon)} \sim \mathbb{M}^{(\mathrm{sc})}_{a_1 \to \pi f_0} |\vec{p}_\pi|.$$

If one fixes mass of f_0, i.e. $p_{f_0}^2 = m_{f_0}^2$, then only $p_0^2 = s$ is variable.

VPP and SSS



+ finite width of $K^\star$

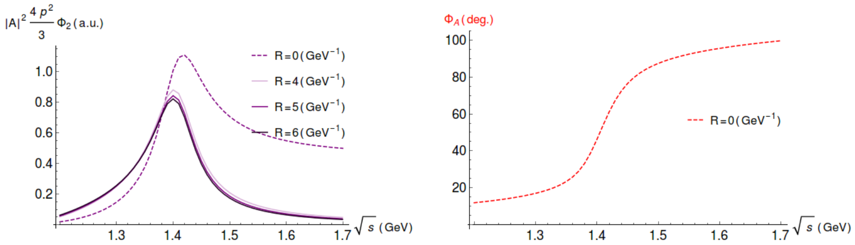


Further corrections to the amplitude

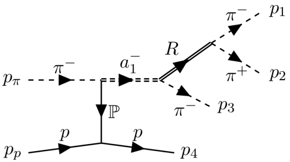
$K^\star(892) \to K\pi$, P-wave decay gives tail to the amplitude.
A left-hand singularity is introduced to correct the amplitude
$g_{K^\star K\pi} \to g_{K^\star K\pi} \times F(k_1)$. k_1 is $K^\star$ four-momentum.

$$F(k_1) = \frac{M^2 - m_{K^\star}^2}{M^2 - k_1^2}, \quad M^2 = (m_\pi + m_K)^2 - \frac{4}{R^2} \ .$$

M is position of the left singularity, it corresponds to the size of $K^\star$:
$F \approx (1 + R^2|\vec{p_0}\,|^2)/(1 + R^2|\vec{p}\,|^2)$. $|\vec{p}|$ is $K^\star \to K\pi$ break up momentum.



Magnitude of the effect

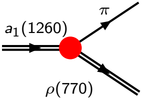


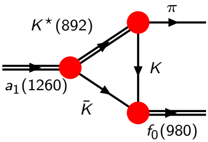
Following the experiment, we compare

- main channel $a_1(1260) \to \rho(770)\pi$ S-wave decay,
- channel $a_1(1260) \to f_0(980)\pi$ only via kaon rescattering

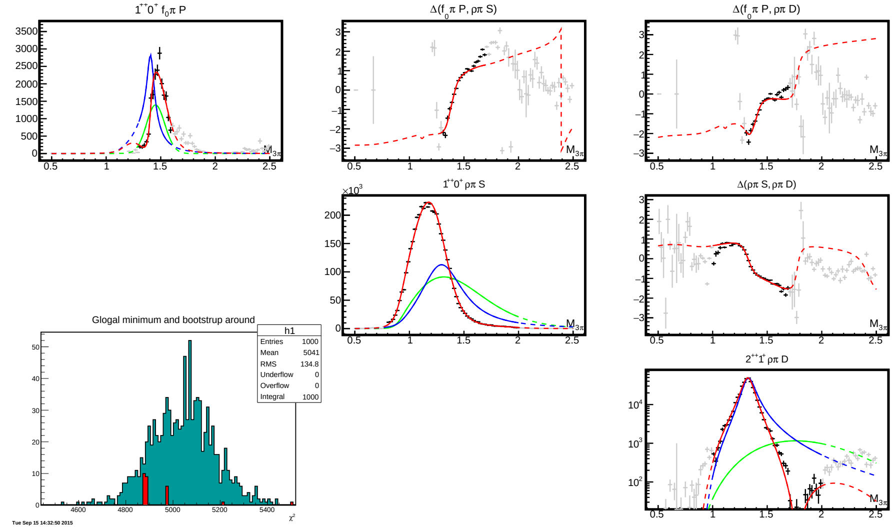
Couplings: $a_1(1260) \to K^\star K, \rho\pi$, ($\tau-$decay) $K^\star \to K\pi$, ($K^\star-$decay) $f_0 \to K\bar{K}$. (General analysis of $\pi\pi, R_{K\bar{K}/\pi\pi}$)

[Coan *et al.*, Phys.Rev.Lett. **92**, 232001 (2004)] [Garcia-Martin *et al.*, Phys.Rev.Lett. **107**, 072001 (2011)]

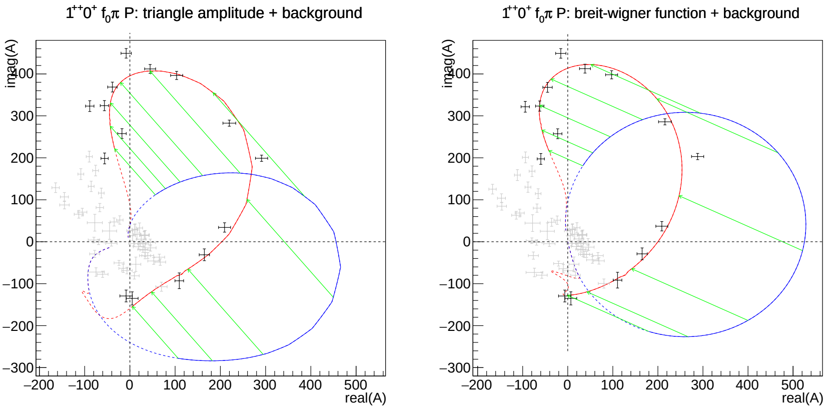




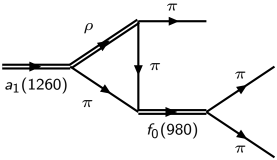
Fit the amplitude $M_{\Delta}^{(sc)}$ to the COMPASS data

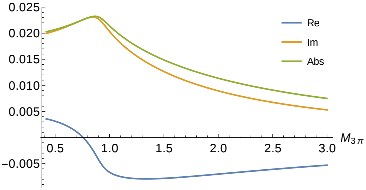


Fit to the data



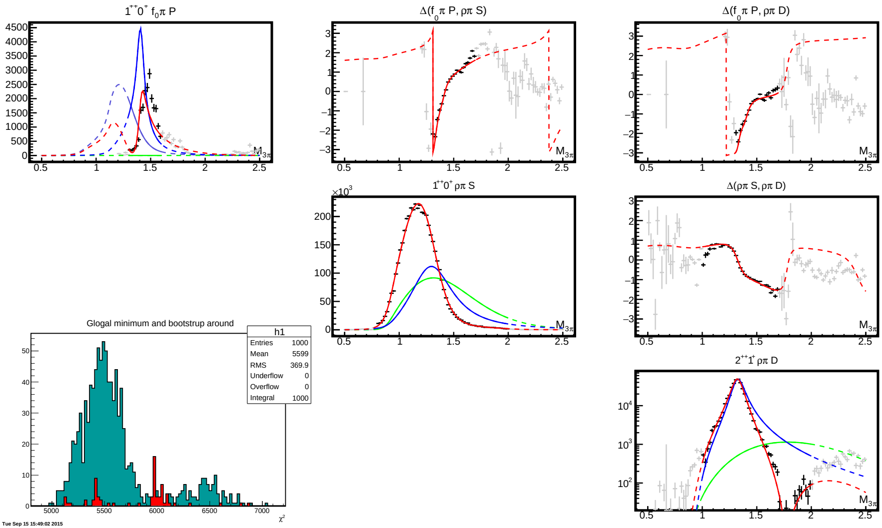
$\rho\pi \to f_0\pi$ rescattering as a background





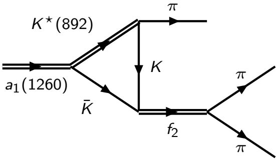
- $a_1(1260) \to \rho\pi$ is the main channel of decay,
- the logarithmic singularity from triangle is far at the complex plane,
- the phase in the physical region is almost const,
- it produce a smoth "background".

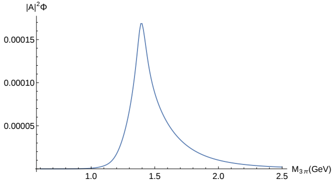
Fit with two triangle amplitude

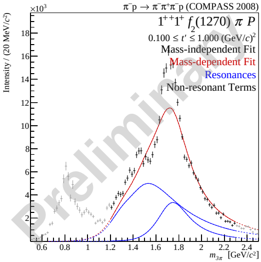


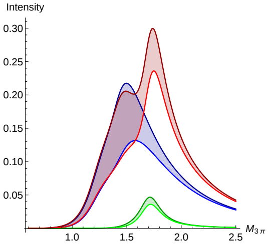
1^{++} $f_2\pi$ P-wave

- The singularity is not in the physical region, but rather close.
- The threshold for $f_2\pi$ opens at $\sim 1.4\,\mathrm{GeV}$.









Local conclusion-III

- The rescattering effects becomes very important when additioonal channels open and inelastisity rises.
- The rescattering causes the wide smooth contributions to the waves and in some paricular case can produce resonance-like peaks.
- $X \to K^\star K \to \pi Y_{(K\bar{K})}$ is ideal kinematics to produce narrow bump and sharp phase motion, which can be recognised in data.
- $K^\star K \to f_0\pi$ rescattering makes the resonance-like signal $a_1(1420)$ at 1^{++} $f_0\pi P - wave$. The fit is reasonable and comparable with the Briet-Wigner fit.
- Analysis of $a_1 \to KK\pi$ is needed.

Conclusion

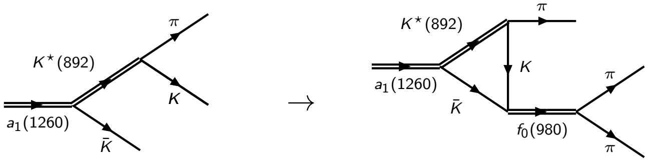
- Rescattering in the triangle diagram is an important mechanism in hadron interactions.
  - Simplest diagram to produce dynamic singularity which gives infinity.
  - Triangle singularity appears not only as anomalous threshold, but also as additional threshold above normal one in decay kinematics.
  - The position of singularity in the decay kinematics is given by simple condition of rescattering of real particles.
- The full rescattering amplitude in presence of many inelastic channels makes the calculations more challenging. The experimental data on channels which rescatter to each others are needed.
- We have now several hints that the triangle singularity plays important role in meson spectroscopy.

Thank you for attention

Backup

The interpretations of $a_1(1420)$

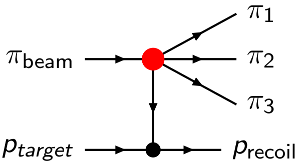
- 4-quark state candidate [Hua-Xing Chen *et al.*, arXiv:1503.02597], [Zhi-Gang Wang, arXiv:1401.1134].
- $K^\star K$ molecule (similar to XYZ interpretation)
- Dynamic effect of interference with Deck [Basdevant *et al.*, arXiv:1501.04643] .
- Triangle singularity [Mikhasenko *et al.*, arXiv:1501.07023] :

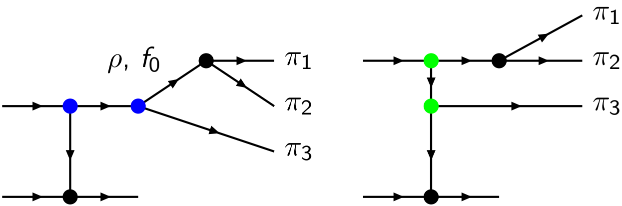


final state rescattering of almost real particle. Logarithmic singularity in the amplitude of the processes:

Reactions at fixed-target experiments

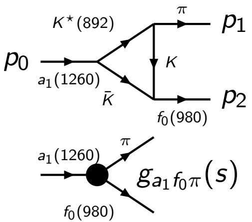
- COMPASS (VES) is a fixed-target experiment.
- 190 GeV (29 GeV) pion beam.
- The recoil proton gives trigger (veto).
- Charged particles are observed by spectrometer, neutral ones are registered using calorimeters.

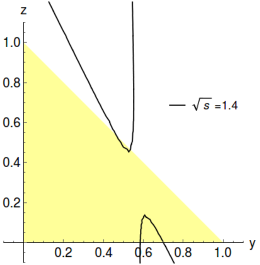


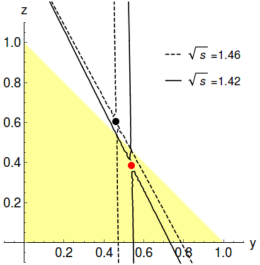


Resonance production and Deck-production amplitude.

The pinch singularity







$$g_{a_1 f_0 \pi}(s|_{s=p_0^2}) = g^3 \int \frac{d^4 k_1}{(2\pi)^4 i} \frac{1}{\Delta_1 \Delta_2 \Delta_3} = \frac{g^3}{16\pi^2} \int_0^1 \mathrm{d}y \int_0^{1-y} \mathrm{d}z \, \frac{1}{D},$$

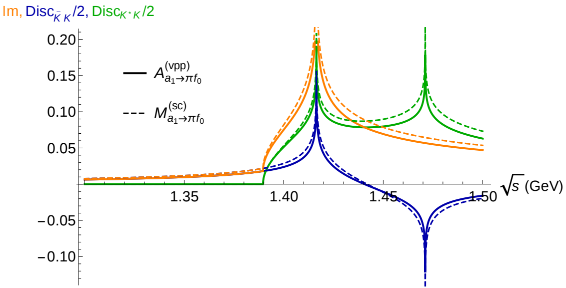
$$\Delta_i = m_i^2 - k_i^2 - i\epsilon, \quad D = (1-y-z)m_1^2 + y m_2^2 + z m_3^2 - y(1-y-z)p_0^2 - z(1-z-y)p_1^2 - yz p_2^2.$$

If Landau conditions [Nucl. Phys. **13**, 181 (1959)] are satisfied, $g_{a_1 f_0 \pi} \sim \log(s - s_0)$.

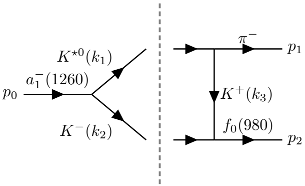
For a **box loop**, $A \sim (s - s0)^{-1/2}$, for **5-leg loop** $A \sim (s - s_0)^{-1}$ (pole).

The imaginary part based on Cutkosky cutting rules

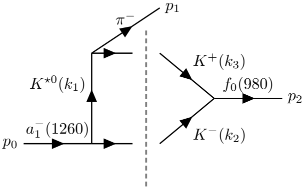
$$2\mathrm{Im}A = \mathrm{Disc}_{K\star\bar{K}}\, A + \mathrm{Disc}_{K\bar{K}}\, A$$



$\mathrm{Disc}_{K\star\bar{K}}\, A$:



$\mathrm{Disc}_{K\bar{K}}\, A$:

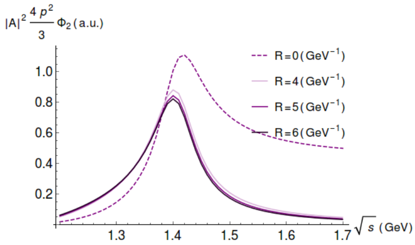


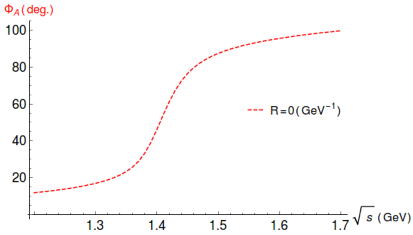
SP-corrections to the amplitude

$K^\star(892) \to K\pi$, P-wave decay gives tail to the amplitude.
A left-hand singularity is introduce to correct the amplitude
$g_{K^\star K\pi} \to g_{K^\star K\pi} \times F(k_1)$. k_1 is $K^\star$ four-momentum.

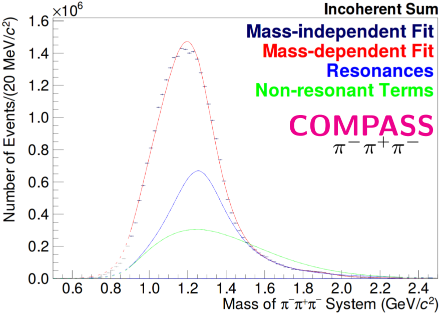
$$F(k_1) = \frac{M^2 - m_{K^\star}^2}{M^2 - k_1^2}, \quad M^2 = (m_\pi + m_K)^2 - \frac{4}{R^2} \ .$$

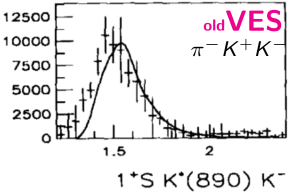
M is position of the left singularity, it corresponds to the size of $K^\star$:
$F \approx (1 + R^2|\vec{p_0}|^2)/(1 + R^2|\vec{p}|^2)$. $|\vec{p}|$ is $K^\star \to K\pi$ break up momentum.





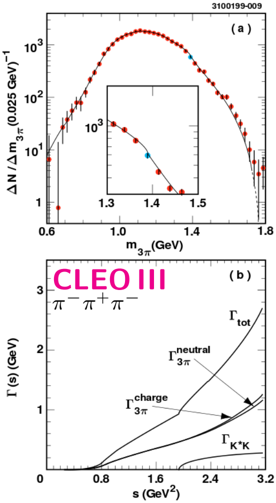
$a_1(1260)$ measurements





[Berdnikov *et al.*, Nuovo Cim. 107, 1941 (1994)]

[Aster *et al.*, Phys.Rev. D, **60**, 0120002 (1999)]



List of $a_1(1260)$ decays

TABLE III. Results of the nominal fit for the moduli $|\beta_i|$ and phases ϕ_{β_i} of the coefficients for the amplitudes listed in Eq. (6). The two errors shown are statistical and systematic, respectively. The branching fractions $\mathcal{B}$ are derived from the squared amplitudes (using the values of $|\beta_i|$), and are normalized to the total $\tau^- \to \nu_\tau \pi^- \pi^0 \pi^0$ rate. These do not sum to 100%, due to interference between the amplitudes.

		Signif.	$\lvert\beta_i\rvert$	ϕ_{β_i}/π	$\mathcal{B}$ fraction (%)
ρ	s-wave		1	0	68.11
$\rho(1450)$	s-wave	1.4σ	$0.12\pm0.09\pm0.03$	$0.99\pm0.25\pm0.04$	$0.30\pm0.64\pm0.17$
ρ	d-wave	5.0σ	$0.37\pm0.09\pm0.03$	$-0.15\pm0.10\pm0.03$	$0.36\pm0.17\pm0.06$
$\rho(1450)$	d-wave	3.1σ	$0.87\pm0.29\pm0.06$	$0.53\pm0.16\pm0.06$	$0.43\pm0.28\pm0.06$
$f_2(1270)$	p-wave	4.2σ	$0.71\pm0.16\pm0.05$	$0.56\pm0.10\pm0.03$	$0.14\pm0.06\pm0.02$
σ	p-wave	8.2σ	$2.10\pm0.27\pm0.09$	$0.23\pm0.03\pm0.02$	$16.18\pm3.85\pm1.28$
$f_0(1370)$	p-wave	5.4σ	$0.77\pm0.14\pm0.05$	$-0.54\pm0.06\pm0.02$	$4.29\pm2.29\pm0.73$

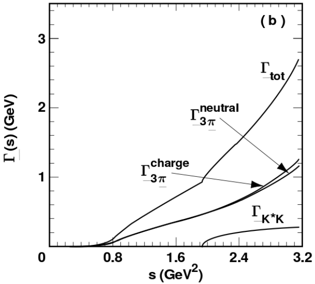
[Aster *et al.*, Phys.Rev. D, **60**, 0120002 (1999)]

a_1 decays, [Aster *et al.*, Phys.Rev. D, **60**, 0120002 (1999)]

The best knowledge is from τ decays (mass dependent analysis).
Short results: $m_{a_1} \approx 1.33\,\text{GeV}$, $\Gamma_{a_1}(m_{a_1}) \approx 814\,\text{GeV}$,
$\text{Br}(a_1 \to \rho\pi) \approx 60\%$, $\text{Br}(a_1 \to \sigma\pi) \approx 20\%$, $\text{Br}(a_1 \to f_0(1370)\pi) \approx 7\%$,
$\text{Br}(a_1 \to K^\star \bar{K}) \approx 2.2\%$.

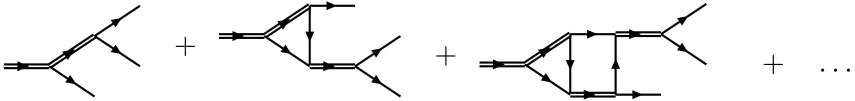
Correction for 'finite size of meson' is done by exponential suppression (not by Blatt-Weitskopf), $R = 1..2/GeV^2$ is strongly correlated with $\Gamma_{a_1}(m_{a_1})$.

- $a_1 \to \rho\pi$ S wave,
- $a_1 \to \sigma\pi$ P wave,
- $a_1 \to K^\star \bar{K}$ S wave.



Let final states re-scatter

Isobar model $+$ 1$^{\text{th}}$-rescattering $+$ 2$^{\text{th}}$-rescattering $+$ etc.



Possible isobars from $a_1(1260)$: $\rho(770)$, $f_0(500)$, $\rho(1450)$, $f_0(1370)$, $K^\star(892)$. And then all possible particles can be result of secondary interaction.

$K^\star \to K\pi$ P-wave decay

- P-wave. In our parametrization:

$$\Gamma = \frac{1}{2m_{K^\star}} \frac{1}{8\pi} \frac{2|\vec{k}|}{m_{K^\star}} \times g^2_{K^\star K\pi} \frac{4}{3} |\vec{k}^2|.$$

- $\Gamma_{K^\star \to K\pi} = 50.8\,\mathrm{MeV}$. So, $g^2_{K^\star K\pi} = 31.2$.
- From isospin $\mathrm{Br}(K^{\star -} \to K^0\pi^-)/\mathrm{Br}(K^{*-} \to K^-\pi^0) = 2$.

$f_0(980)$ decays

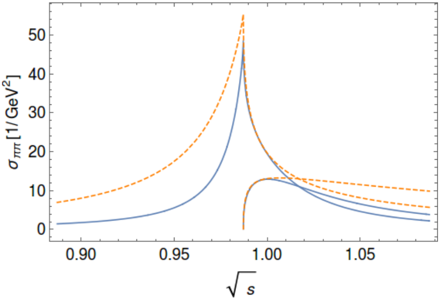
- Flatte Flatte, Phys.Lett.(1976),
- Flatte-Like Achasov(2003), Bary(2005),

The problem with parameters of f_0 is scaling behaviour near $K\bar{K}$ threshold. It results in $R = g_{KK}^2/g_{\pi\pi}^2$ is known (more or less) $\approx 3\dots 4$, branching ratio $f_0 \to \pi\pi$ is ≈ 0.4, but g_{KK}, $g_{\pi\pi}$ variates.

$$\bar{g}_{\pi\pi} = \frac{g_{\pi\pi}^2}{8\pi m_{f_0}^2} = 0.1\dots 0.6$$

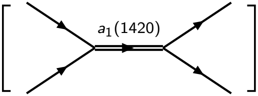
$$f_e \sim \frac{1}{m_{f_0}^2 - s - i(g_{\pi\pi}^2\rho_{\pi\pi} + g_{KK}^2\rho_{KK})/2}$$

$$f_e \sim \frac{1}{m_{f_0}^2 - s - im_{f_0}(\bar{g}_{\pi\pi}k_\pi + \bar{g}_{KK}k_K)}$$



What is a resonance?

$$A = \left[\ \right] \sim \frac{1}{m^2 - s - i\, m\, \Gamma_{\mathrm{tot}}}$$



A resonance is a pole in the amplitude.
Manifestation:

- Bump in $\sqrt{s} = M_{12}$ with finite width.
- Phase motion. Only relative phase motion is observed.